



Liu, Yu-shan

Contact

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Profile

My name is Liu, Yu-shan, and I am currently pursuing a master's degree at NCKU. I am a fast learner with strong analytical, problem-solving, and stress-management skills. Through academic projects, extracurricular activities, and investment research, I have developed the ability to solve problems through analysis, implementation, and continuous validation.

My academic focus is machine learning and data analysis, and I am proficient in C, C++, and Python with hands-on project experience. I am also passionate about equity research and industry analysis, applying fundamental analysis and data-driven research to evaluate business prospects and investment opportunities.

Work Experience

Macronix | 06/2022 - 04/2024

Diffusion Equipment Engineer

- Fab-Wide Equipment Performance Improvement Program
 - Improved the yield of a key product by 2%, generating approximately NT\$10 million in annual cost savings, while maintaining monthly equipment utilization at 90% or above.
- Developed a quartz consumables recycling and cost optimization program
 - Reducing monthly procurement expenses by approximately NT\$300,000 (15%).

Education

National Cheng Kung University, M.S. in Department of Electrical Engineering | 2025/06 - Present

Machine Learning-Based Metabolite Screening Platform | 2025

- Developed a web-based platform that applies machine learning to identify metabolites matching user-defined criteria.
- Prioritized candidate metabolites potentially associated with cancer from a large pool of unvalidated data.
- Reduced the scope, time, and cost of downstream experimental validation.

UAV Semantic Segmentation Competition | 2025

- Developed a deep learning semantic segmentation model to classify objects in UAV aerial images, including trees, roads, buildings, and vehicles.
- 4th on the private leaderboard among 69 teams.

TMBA, ECM | 2026/07 - Present

- Packaging & Testing Team - ASE Technology Equity Research

Tamkang University, B.S. in Aerospace Engineering | 2017/09 - 2021/06

Solar Power Monitoring System for a Green-Energy UAV, Project | 2020

- Developed a system to monitor and analyze UAV power consumption during flight.
- Determined optimal switching between solar and battery power to improve energy efficiency.

Mountain Club, Chief Guide & Records Officer | 2020/09 - 2021/06

- Led four trekking trips, overseeing pre-trip planning, team coordination, and emergency response.
- Designed and delivered training in mountaineering skills, preparing junior members to lead trips independently.

Side Project

Equity Research - APT(7734.TW)

<https://reurl.cc/p8WWkb>

Machine Learning-Based Metabolite Screening Platform

https://cosbi10.ee.ncku.edu.tw/DEM_screener/